F4. ARUSHA DC MOCK EXAM

MARKING GUIDE

FOR ADDITIONAL MATHEMATICS

1. (a) Let T be the number of Tractors
A be the number of acres cultivated.

$T \propto A$

$T = kA$ — ($0\frac{1}{2}$)

$k = \frac{T}{A}$

$= \frac{2}{20}$ — ($0\frac{1}{2}$)

$= \frac{1}{10}$

$T = \frac{1}{10}A$ — (01)

$5 = \frac{1}{10}A$

$A = 5 \times 10$

$= 50$ — (01)

∴ When 3 tractors of the same make are added 50 acres of land will be cultivated.

(b) $m \propto \frac{b}{L}$

$m = \frac{kb}{L}$

$20 = \frac{2k}{15}$ — ($0\frac{1}{2}$)

$k = \frac{20 \times 15}{2} = 150$

1. (b) $m = 150\frac{b}{L}$

$= \frac{150 \times 3}{20}$

$= 22.5$ kg — $(01\frac{1}{2})$

2. (a) The given data can be arranged as

10, 11, 15, 19, 20, 24, 28, 34, 37, 47, 50, 57

Q_1 Q_2 Q_3

Lower quartile $= \frac{15+19}{2}$

$= 17$ — $(01\frac{1}{2})$

Upper quartile $= \frac{37+47}{2}$

$= 42$ — $(01\frac{1}{2})$

(b) mean $= A + \frac{\sum d}{N}$, let $A = 28$ — (01)

$\bar{x} = 28 + \frac{(10-28)+(11-28)+(15-28)+(19-28)+(20-28)+(24-28)+(34-28)+(37-28)+(47-28)+(50-28)+(57-28)}{12}$ — (01)

$= 29.33$ — (01)

3. (a) The slope of a line through Q and R $= \frac{6-5}{-3-2}$

$= -\frac{1}{5}$

The slope of line through P $= 5$ $(m_1 m_2 = -1)$

Eqn of the line through P is $y = 5(x+3)+5$

$y = 5x + 20$ — (01)

from $(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

$= \left(\frac{-3m+2n}{m+n}, \frac{6m+5n}{m+n}\right)$ — (00½)

Since the line $y = 5x + 20$ pass through (x, y), it satisfies.

That is $\frac{6m+5n}{m+n} = 5\left(\frac{-3m+2n}{m+n}\right) + 20$

$= \frac{-15m + 10n + 20m + 20n}{m+n}$ (00½)

$6m + 5n = 5m + 30n$

$m = 25n$

$\frac{m}{n} = \frac{25}{1}$

$\therefore m : n = 25 : 1.$ — (01)

3. (b) Required to find eqn of a circle with centre at $(4,-7)$ and touches the line $3x+4y-9=0$

From $d = \left| \frac{ax+by+c}{\sqrt{a^2+b^2}} \right|$

$$r = \left| \frac{3(4)+4(-7)-9}{\sqrt{3^2+4^2}} \right|$$

$$= \left| \frac{12-28-9}{\sqrt{25}} \right| = \left| -\frac{25}{5} \right| = 5$$

$= 5$ units — $(01\frac{1}{2})$

From $(x-a)^2+(y-b)^2 = r^2$

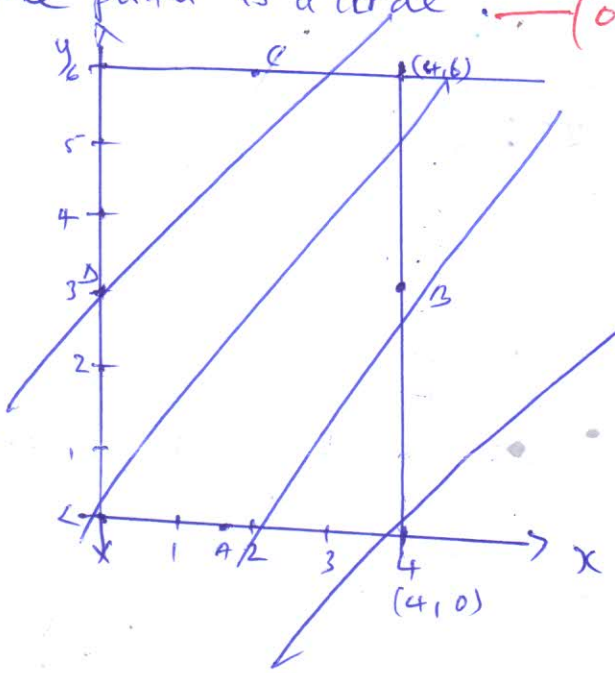
$(x-4)^2+(y--7)^2 = 5^2$

$x^2-8x+16+y^2+14y+49 = 25$ — $(01\frac{1}{2})$

$x^2+y^2-8x+14y+40 = 0$, This is the required eqn of the circle

4. (a) The path is a circle — (01)

(b)





4(b) $4x^2 + 4y^2 + 4x + 16y - 47 = 0$

$x^2 + y^2 + x + 4y = \frac{47}{4}$

$x^2 + x + \frac{1}{4} + y^2 + 4y + 4 = 4 + \frac{1}{4} + \frac{47}{4}$ —(01)

$\left(x + \frac{1}{2}\right)^2 + (y+2)^2 = 16$ —— (01)

$\left(x - -\frac{1}{2}\right)^2 + (y - -2)^2 = 4^2$ —— (01)

(i) The coordinates of Q are $(-\frac{1}{2}, -2)$ —— (01)

(ii) The constant distance is 4 units —— (01)

5. (a) Let x be the number of bottles of pepsi of cost of Tzs 600 each.

Then $500y = 600x$ —— (00½)

$y - x = 1$ —— (00½)

$\Rightarrow y = x + 1$

$500(x+1) = 600x$ —— (00½)

$500x + 500 = 600x$

$100x = 500$ —— (01)

$x = 5$

∴ There are 5 bottles of pepsi which cost Tzs 600 each. —— (00½)

5. (b) Let x and y be the cost of each pen and exercise book, respectively.

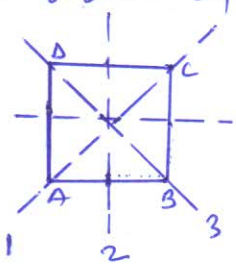
Then $\begin{cases} 4x + 5y = 4800 & \text{— (i)} \\ 12y - 10x = 7600 & \text{— (ii)} \end{cases}$ — (00½) — (00½)

on solving eqn (i) and (ii), simultaneously,

we get $x = 200$ — (01)

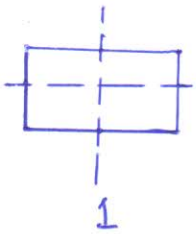
$y = 800$ — (01)

6. (a) - 4 lines of symmetry
- 4 order of rotational symmetry } (00½) (a)



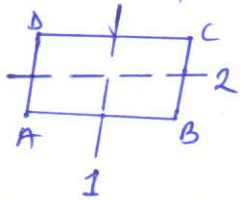
} (00½)

(b) - 2 lines of symmetry
- 2 order of rotational symmetry } (00½) (a)



} (00½)

(c) - 2 lines of symmetry
- 2 order of rotational symmetry } (00½) (a)



} (00½)

(d) - Infinite lines of symmetry
- uncountable order of rotational symmetry } (00½) (a)



} (00½)

7. (a) $\cos 2\theta + \cos\theta = 0$

$\cos^2\theta - \sin^2\theta + \cos\theta = 0$

$\cos^2\theta - (1 - \cos^2\theta) + \cos\theta = 0$ ——— (00½)

$2\cos^2\theta + \cos\theta - 1 = 0$

on solving $\cos\theta = \frac{1}{2}$, $\cos\theta = -1$ ——— (00½)

For $\cos\theta = \frac{1}{2}$

$\theta = \cos^{-1}\left[\frac{1}{2}\right]$

$= 60°, 300°$ ——— (01)

For $\cos\theta = -1$

$\theta = \cos^{-1}[-1]$

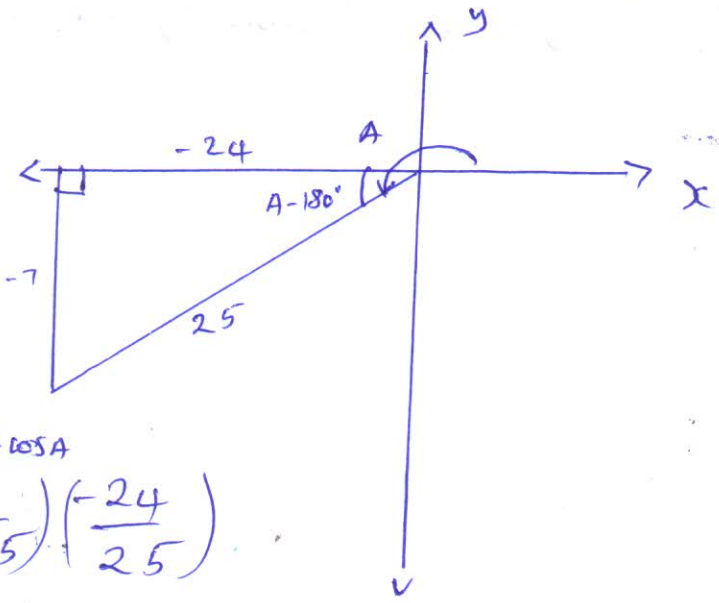
$= 180°$ ——— (00½)

$\therefore \theta = 60°, 180°$ and $300°$ ——— (00½)

(b) $\tan A = \frac{7}{24}$

$\sin A = -\frac{7}{25}$

$\cos A = -\frac{24}{25}$



(i) $\sin 2A = 2\sin A\cos A$

$= 2\left(-\frac{7}{25}\right)\left(-\frac{24}{25}\right)$

$= \frac{336}{625}$ ——— (01)

(ii) $\cos 2A = \cos^2 A - \sin^2 A$

$= \left(-\frac{24}{25}\right)^2 - \left(-\frac{7}{25}\right)^2$

$= \frac{527}{625}$ ——— (01)

7. (b) (iii) $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

$$= \frac{2\left(\frac{7}{24}\right)}{1-\left(\frac{7}{24}\right)^2}$$

$$= \frac{336}{527}$$ —— (01)

8. (a) The number is divisible by 9 if the sum of digits is divisible by 9 —(01)

12357; $1+2+3+5+7 = 18$, which is divisible by 9 —(01)

$\therefore$ 12357 is divisible by 9. —(01)

(00½)

8. (b) (ii) 50 and 59 —— (00½)

(ii) $A_6 = A_4 + A_5$ $A_7 = A_5 + A_6$

$= 90$, $= 56 + 90$

(01) $= 146$ —— (01)

9. (a) Let p be clouds are heavy

q be it rain

The argument symbolically is written as

$[(p \to q) \wedge \sim q] \to \sim p$ —— (01)

using the truth table

p	q	$\sim p$	$\sim q$	$p \to q$	$(p \to q) \wedge \sim q$	$((p \to q) \wedge \sim q) \to \sim p$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	F	T
F	F	T	T	T	T	T

The argument end with T only, hence (00½) it is valid. — 00½

(b) (i) Let p be Erick reads newsweek

q be life

r be Time

(i) $(p \vee q) \wedge \sim r$ —— (01)

(ii) $(p \wedge q) \vee (p \wedge r)$ —— (01)

(iii) $\sim(p \wedge \sim r)$ —— (01)

10. (a) (i) $B \cap C = \{4, 5\}$ —— 00½

$A \cup (B \cap C) = \{1, 2, 3, 4, 5\}$ —— (00½)

(ii) $B \cup C = \{3, 4, 5, 6, 7\}$ —— (00½)

$A \cap (B \cup C) = \{3\}$ —— (00½)

11. (a) Let $f(x) = ax^3 + x^2 + bx + 6$

$x - 2 = 0$

$x = 2$

$f(2) = 0$

$a(2^3) + 2^2 + 2b + 6 = 0$

$8a + 2b = -10$ —(i) —— (01)

$x + 4 = 0$

$x = -4$.

$f(-4) = 10$

$a(-4)^3 + (-4)^2 - 4b + 6 = 10$

$-64a - 4b = -12$ —(ii) —— (01)

Solving (i) and (ii), simultaneously, we get

$a = 2/3$ —— (01)

$b = -\frac{23}{3}$ —— (01)

(b) $x^2 + 3x + 4 = 0$

$\alpha + \beta = -3$ —— (00½)

$\alpha\beta = 4$ —— (00½)

(i) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= (-3)^3 - 3(4)(-3)$

$= -27 + 36$

$= 9$ —— (01)

$\alpha^3\beta^3 = (\alpha\beta)^3$

$= 4^3 = 64$ —— (01)

11. (b) From $x^2 - (\text{Sum of roots})x + \text{product of roots} = 0$

$x^2 - 9x + 64 = 0$ ——— (01)

(ii) $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{\alpha^3 + \beta^3}{(\alpha\beta)^3}$

$= \frac{9}{64}$ ——— (00½)

$\frac{1}{\alpha^3} \times \frac{1}{\beta^3} = \frac{1}{64}$ ——— (00½)

Hence, $x^2 - \frac{9}{64}x + \frac{1}{64} = 0$

$\Rightarrow 64x^2 - 9x + 1 = 0$ ——— (01)

12 (a) Let A be the surface area of the sphere with radius r, then

$A = 4\pi r^2$ --- ($00\frac{1}{2}$)

$\frac{dA}{dr} = 8\pi r$ and $\partial A = 8\pi r \partial r$ --- (01)

but $\partial r = 0.02r$

$\partial A = 8\pi r\left(\frac{2}{100}r\right)$

$= \frac{16\pi r^2}{100}$ --- (01)

$\frac{\partial A}{A} = \frac{16\pi r^2}{100} \times \frac{1}{4\pi r^2} = \frac{4}{100}$ --- ($00\frac{1}{2}$)

Hence, percentage area in surface area $= 4\%$ --- (01)

(b) Given $y = x^2 - 4x - 12$, At y-axis $x = 0$

when $x = 0$, $y = 0^2 - 4(0) - 12 = -12$

$\frac{dy}{dx} = 2x - 4$ --- (01)

$\frac{dy}{dx}(x=0) = 2(0) - 4$

$= -4$, This is the slope of the tangent, which is also the slope of the curve at $(0, -12)$.

from $m_N m_T = -1$

$m_N = \frac{-1}{-4} = \frac{1}{4}$. --- (01)

Equation of normal

$\frac{1}{4} = \frac{y+12}{x-0}$

$4y - x = 48$; This is the required eqn of the normal line. --- (01)

12. (c) Given $y = 4x - x^2$, $x = 1$, $x = 2$ and the x-axis

From Area $= \int_a^b y\,dx$

$= \int_1^2 (4x - x^2)\,dx$ — — — (01)

$= \left[4\frac{x^2}{2} - \frac{x^3}{3}\right]_1^2$ — — — (01)

$= \left(2(2^2) - \frac{2^3}{3}\right) - \left(2(1^2) - \frac{1^3}{3}\right)$

$= \frac{16}{3} - \frac{5}{3}$

$= \frac{11}{3}$ or 3.67 square units — — — (01)

13. (a) Given 5 children

(i) For two of them to together, we consider them as 1 child and they arranged themselve in 2! ways. — — — (01)

Remained 4 children, they can be arranged in 4! ways. — — — (01)

Hence, the required number of arrangements $= 4!\,2!$

$= 48$ — — — (01)

(ii) Total number of arrangements of 5 children $= 5!$

$= 120$ — (01)

∴ Number of arrangements for Amit and Ajit are never together

$= 120 - 48$

$= 72$ — — — (01)

(b) $P(A \cap B) = P(A) \times P(B)$ for A and B to be independent events. — (01)

$P(\bar{A} \cap \bar{B}) = P(A \cup B)'$

$= 1 - P(A \cup B)$ — — — (01)

$= 1 - [P(A) + P(B) - P(A \cap B)]$ — — (01)

13. (b) $= 1 - [P(A) + P(B) - P(A) \cdot P(B)]$

$= 1 - [P(A) + P(B)[1 - P(A)]]$

$= 1 - P(A) - P(B)(1 - P(A))$ — — — (01)

$= [1 - P(A)](1 - P(B))$ — — — (01)

$= P(\bar{A}) \cdot P(\bar{B})$, hence proved — — — (00 $\frac{1}{2}$)

14. (a) From $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}| \cos\theta$ but $|\underset{\sim}{a}| = |\underset{\sim}{b}|$ and $\theta = 60°$

$\underset{\sim}{a} \cdot \underset{\sim}{b} = \frac{1}{2}$.

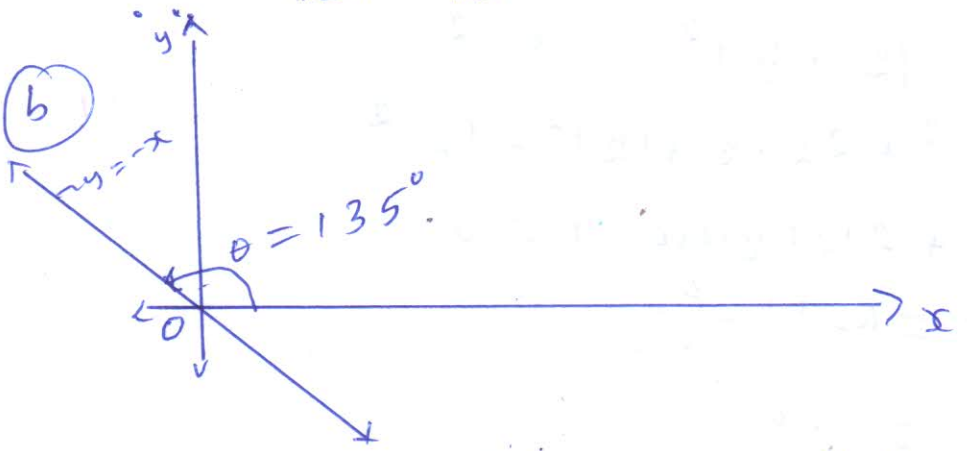
$\frac{1}{2} = |\underset{\sim}{a}|^2 \cos 60°$ — — — (00 $\frac{1}{2}$)

$= \frac{1}{2}|\underset{\sim}{a}|^2$

$|\underset{\sim}{a}|^2 = 1$

$|\underset{\sim}{a}| = |\underset{\sim}{b}| = 1$ — — — (01)

(b)



(00 $\frac{1}{2}$)

From $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$= \begin{pmatrix} \cos 270° & \sin 270° \\ \sin 270° & -\cos 270° \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ — — (01)

13. (b) $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$= \begin{pmatrix} -4 \\ -3 \end{pmatrix}$ — — — (01)

The line $y = 0$, $\theta = 180°$ — — — (00½)

Then, $\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} \cos 360° & \sin 360° \\ \sin 360° & -\cos 360° \end{pmatrix} \begin{pmatrix} -4 \\ -3 \end{pmatrix}$ — — — (01)

$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -4 \\ -3 \end{pmatrix}$

$= \begin{pmatrix} -4 \\ 3 \end{pmatrix}$ — — — (01)

(c) Given $\underset{\sim}{a} + \underset{\sim}{b} = -\underset{\sim}{c}$, $|\underset{\sim}{a}| = 1$, $|\underset{\sim}{b}| = 3$ $|\underset{\sim}{c}| = 4$.

we have $|\underset{\sim}{a} + \underset{\sim}{b}|^2 = |-\underset{\sim}{c}|^2$

$|\underset{\sim}{a}|^2 + 2\,\underset{\sim}{a} \cdot \underset{\sim}{b} + |\underset{\sim}{b}|^2 = |\underset{\sim}{c}|^2$ — — — (01)

$1 + 2|\underset{\sim}{a}||\underset{\sim}{b}|\cos\theta + 9 = 16$ — — — (01)

$\theta = \cos^{-1}\left[\frac{3}{1 \times 3}\right]$

$= 0°$

$\therefore$ The angle btw $\underset{\sim}{a}$ and $\underset{\sim}{b}$ is $0°$ — — — (01)